

Example 18. *Let*

$$s_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}, \quad n \in \mathbb{N}.$$

We show, using the Cauchy Criterion, that (s_n) is not convergent.

Proof. By the Cauchy Criterion, it is enough to show that (s_n) is not a Cauchy sequence. Choose $\varepsilon = \frac{1}{2}$ and let $N \in \mathbb{N}$ be arbitrary. Take $n \geq N$ and let $m = 2n$. Then

$$\begin{aligned} |s_m - s_n| &= |s_{2n} - s_n| = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \\ &\geq \underbrace{\frac{1}{2n} + \frac{1}{2n} + \cdots + \frac{1}{2n}}_{n \text{ terms}} = \frac{1}{2}. \end{aligned}$$

Thus, for every $N \in \mathbb{N}$, there exist $m, n \geq N$ such that

$$|s_m - s_n| \geq \frac{1}{2}.$$

Hence, (s_n) is not a Cauchy sequence. Therefore, by the Cauchy Criterion, (s_n) is not convergent. $\square$

Example 19. *The sequence $((-1)^n)$ is not a Cauchy sequence.*

Proof. Choose $\varepsilon = 1$. Let $N \in \mathbb{N}$ be arbitrary. Take

$$m = \begin{cases} N+1, & \text{if } N \text{ is even,} \\ N+2, & \text{if } N \text{ is odd,} \end{cases} \quad \text{and} \quad n = m+1.$$

Then $m, n \geq N$, $(-1)^m = 1$, and $(-1)^n = -1$. Hence,

$$|(-1)^m - (-1)^n| = |1 - (-1)| = 2 > 1 = \varepsilon.$$

Therefore, $((-1)^n)$ is not a Cauchy sequence. $\square$

2.4 Monotone sequences

Definition 12. *A sequence (x_n) is said to be increasing if*

$$x_n \leq x_{n+1}, \quad \forall n \in \mathbb{N}.$$

Definition 13. A sequence (x_n) is said to be strictly increasing if

$$x_n < x_{n+1}, \quad \forall n \in \mathbb{N}.$$

Definition 14. A sequence (x_n) is said to be decreasing if

$$x_n \geq x_{n+1}, \quad \forall n \in \mathbb{N}.$$

Definition 15. A sequence (x_n) is said to be strictly decreasing if

$$x_n > x_{n+1}, \quad \forall n \in \mathbb{N}.$$

Definition 16. A sequence is said to be monotone if it is either increasing or decreasing.

Remark 2. Every increasing sequence is bounded below by x_1 . Hence, an increasing sequence is bounded if and only if it is bounded above. Similarly, every decreasing sequence is bounded above by x_1 . Therefore, a decreasing sequence is bounded if and only if it is bounded below.

Example 20. The following are examples of monotone sequences.

1. The constant sequence (c) is both increasing and decreasing.
2. $(\frac{1}{n})$ is strictly decreasing.
3. (n) is strictly increasing.
4. (2^n) is strictly increasing.
5. $(\frac{n^2+1}{n^2}) = (1 + \frac{1}{n^2})$ is strictly decreasing.
6. $((-1)^n)$ is not monotone.

Theorem 11 (Monotone Convergence Theorem). An increasing sequence of real numbers is convergent if and only if it is bounded above.

Proof. Let (x_n) be an increasing sequence that is bounded above. We show that (x_n) is convergent.

Let

$$E = \{x_n : n \in \mathbb{N}\}.$$

Since E is nonempty and bounded above, by the completeness property of $\mathbb{R}$, the set

E has a least upper bound. Let

$$\ell = \sup E.$$

Let $\varepsilon > 0$ be given. Since $\ell - \varepsilon$ is not an upper bound of E , there exists $N \in \mathbb{N}$ such that

$$x_N > \ell - \varepsilon.$$

As (x_n) is increasing, for every $n \geq N$,

$$\ell - \varepsilon < x_N \leq x_n \leq \ell < \ell + \varepsilon.$$

Hence,

$$|x_n - \ell| < \varepsilon, \quad \forall n \geq N.$$

Therefore,

$$\lim_{n \rightarrow \infty} x_n = \ell.$$

The converse follows immediately from the fact that every convergent sequence is bounded. $\square$

Exercise 11. Show that a decreasing sequence of real numbers is convergent if and only if it is bounded below.

Notation. The notation

$$x_n \downarrow x$$

means that the sequence (x_n) is decreasing and converges to x . Similarly, the notation

$$x_n \uparrow x$$

means that the sequence (x_n) is increasing and converges to x .

Example 21. Let $0 \leq r < 1$. We show that $\lim_{n \rightarrow \infty} r^n = 0$.

If $r = 0$, then there is nothing to prove. So assume that $0 < r < 1$. Since $r^{n+1} < r^n$ for every $n \in \mathbb{N}$, the sequence (r^n) is decreasing. Moreover, it is bounded below by 0. Hence, by the Monotone Convergence Theorem, (r^n) converges. Let $\lim_{n \rightarrow \infty} r^n = \ell$.

By the Algebra of Convergent Sequences, $\lim_{n \rightarrow \infty} r^{n+1} = r\ell$. On the other hand, r^{n+1} is just the shifted sequence of (r^n) , so $\lim_{n \rightarrow \infty} r^{n+1} = \ell$. By the Uniqueness of Limits, $r\ell = \ell$. Since $0 \leq r < 1$, we have $(1 - r)\ell = 0$, and therefore $\ell = 0$. Hence,

$$\lim_{n \rightarrow \infty} r^n = 0.$$

Example 22. Let $0 \leq r < 1$, and define $s_n := 1 + r + r^2 + \cdots + r^n$.

If $r = 0$, then (s_n) is the constant sequence 1, and hence it is convergent. So assume that $0 < r < 1$. Since $s_{n+1} = s_n + r^{n+1} > s_n$, the sequence (s_n) is increasing.

Moreover, $s_n = \frac{1 - r^{n+1}}{1 - r} \leq \frac{1}{1 - r}$, so (s_n) is bounded above. Hence, by the Monotone Convergence Theorem, (s_n) converges.

Finally, $s_n = \frac{1 - r^{n+1}}{1 - r} = \frac{1}{1 - r} - \frac{r^{n+1}}{1 - r}$. Since $\lim_{n \rightarrow \infty} r^{n+1} = 0$, it follows from the Algebra of Convergent Sequences that

$$\lim_{n \rightarrow \infty} s_n = \frac{1}{1 - r}.$$

2.5 Sandwich theorem

In many situations, we may be able to guess the limit of a sequence but find it difficult to prove the convergence directly. A useful idea is to compare the given sequence with two simpler sequences whose limits are already known. If the given sequence always lies between these two sequences, then it is forced to have the same limit. This idea is known as the *Sandwich Theorem*.

Theorem 12 (Sandwich Theorem). *Let (x_n) , (y_n) and (z_n) be sequences of real numbers such that*

1. $\lim_{n \rightarrow \infty} x_n = \alpha$ and $\lim_{n \rightarrow \infty} y_n = \alpha$;
2. $x_n \leq z_n \leq y_n$ for all $n \in \mathbb{N}$.

Then

$$\lim_{n \rightarrow \infty} z_n = \alpha.$$

Proof. Let $\varepsilon > 0$ be given. Since $x_n \rightarrow \alpha$, there exists $N_1 \in \mathbb{N}$ such that $x_n \in (\alpha - \varepsilon, \alpha + \varepsilon)$ for all $n \geq N_1$. Similarly, there exists $N_2 \in \mathbb{N}$ such that $y_n \in (\alpha - \varepsilon, \alpha + \varepsilon)$ for all $n \geq N_2$.

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$,

$$\alpha - \varepsilon < x_n \leq z_n \leq y_n < \alpha + \varepsilon.$$

Hence,

$$|z_n - \alpha| < \varepsilon, \quad \forall n \geq N.$$

Therefore,

$$\lim_{n \rightarrow \infty} z_n = \alpha.$$

□

Example 23. Define the sequence (z_n) by

$$z_n = \frac{\sin n}{n}, \quad n \in \mathbb{N}.$$

We show that $\lim_{n \rightarrow \infty} z_n = 0$.

Since

$$-1 \leq \sin n \leq 1,$$

we have

$$-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}, \quad \forall n \in \mathbb{N}.$$

We already know that

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \left(-\frac{1}{n}\right) = 0.$$

Hence, by the Sandwich Theorem,

$$\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0.$$

Example 24. Show that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2 + 1}} + \frac{1}{\sqrt{n^2 + 2}} + \cdots + \frac{1}{\sqrt{n^2 + n}} \right) = 1.$$

Proof. Let $a_n := \sum_{k=1}^n \frac{1}{\sqrt{n^2 + k}}$. Since $\frac{1}{\sqrt{n^2 + n}} \leq \frac{1}{\sqrt{n^2 + k}} \leq \frac{1}{\sqrt{n^2 + 1}}$, for every $1 \leq k \leq n$,

we obtain $u_n \leq a_n \leq v_n$, where $u_n := \frac{n}{\sqrt{n^2 + n}}$ and $v_n := \frac{n}{\sqrt{n^2 + 1}}$.

Now, $u_n = \frac{1}{\sqrt{1 + \frac{1}{n}}} \rightarrow 1$, and $v_n = \frac{1}{\sqrt{1 + \frac{1}{n^2}}} \rightarrow 1$. Hence, by the Sandwich

Theorem,

$$\lim_{n \rightarrow \infty} a_n = 1.$$

□

Example 25. Let $a > 0$. We show that

$$\lim_{n \rightarrow \infty} a^{1/n} = 1.$$

Proof. If $a = 1$, there is nothing to prove. Assume first that $a > 1$. Write $a^{1/n} = 1 + x_n$, where $x_n > 0$. Then $a = (1 + x_n)^n$. By Bernoulli's inequality, $(1 + x_n)^n \geq 1 + nx_n$.

Hence, $a \geq 1 + nx_n$, which implies $0 < x_n \leq \frac{a-1}{n}$.

Let $\varepsilon > 0$ be given. By the Archimedean Property, choose $N \in \mathbb{N}$ such that $\frac{a-1}{N} < \varepsilon$.

Then, for every $n \geq N$, $0 < x_n \leq \frac{a-1}{n} \leq \frac{a-1}{N} < \varepsilon$. Therefore, $|a^{1/n} - 1| = x_n < \varepsilon$, and hence $a^{1/n} \rightarrow 1$.

Now suppose that $0 < a < 1$. Then $\frac{1}{a} > 1$, and by the first part, $\left(\frac{1}{a}\right)^{1/n} \rightarrow 1$. Since $a^{1/n} = \frac{1}{(1/a)^{1/n}}$, the Algebra of Convergent Sequences gives $a^{1/n} \rightarrow 1$. $\square$

Example 26. Let $0 < a < b$. We show that

$$\lim_{n \rightarrow \infty} (a^n + b^n)^{1/n} = b.$$

Proof. Since $a^n > 0$, we have $b^n \leq a^n + b^n \leq 2b^n$. Taking the n -th root, we obtain $b \leq (a^n + b^n)^{1/n} \leq 2^{1/n}b$.

We already know that $\lim_{n \rightarrow \infty} 2^{1/n} = 1$. Hence, $\lim_{n \rightarrow \infty} 2^{1/n}b = b$. Therefore, by the Sandwich Theorem,

$$\lim_{n \rightarrow \infty} (a^n + b^n)^{1/n} = b.$$

$\square$

Exercise 12. Let (a_n) be a bounded sequence of real numbers and let (x_n) be a sequence such that $x_n \rightarrow 0$.

1. Show that $\lim_{n \rightarrow \infty} a_n x_n = 0$.

2. Deduce that

$$\lim_{n \rightarrow \infty} r^n = 0, \quad \text{for every } |r| < 1.$$

(Hint. Write $r^n = |r|^n \left(\frac{r}{|r|}\right)^n$ when $r \neq 0$.)

Exercise 13 (Some important limits). Solve the following exercises.

1. Show that $\lim_{n \rightarrow \infty} n^{1/n} = 1$.

2. Let $a > 1$. Show that $\lim_{n \rightarrow \infty} \frac{n}{a^n} = 0$.

3. Let $a \in \mathbb{R}$. Show that $\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$.

4. Let (x_n) be a sequence converging to 0. Define $s_n := \frac{x_1 + x_2 + \cdots + x_n}{n}$. Show that (s_n) also converges to 0.

5. Let $x_n \rightarrow x$ and $y_n \rightarrow y$. Show that $\frac{x_1 y_n + x_2 y_{n-1} + \cdots + x_n y_1}{n} \rightarrow xy$.

Example 27. Consider the sequence

$$a_n := \left(1 + \frac{1}{n}\right)^n, \quad n \in \mathbb{N}.$$

We show that (a_n) is increasing and bounded above. Consequently, it is convergent.

Step 1. (a_n) is increasing.

Consider the $(n+1)$ numbers

$$\underbrace{\frac{n+1}{n}, \frac{n+1}{n}, \dots, \frac{n+1}{n}}_{n \text{ times}}, 1.$$

By the Arithmetic Mean–Geometric Mean inequality,

$$\frac{n \left(\frac{n+1}{n}\right) + 1}{n+1} \geq \left[\left(\frac{n+1}{n}\right)^n\right]^{\frac{1}{n+1}}.$$

Since the arithmetic mean is

$$\frac{(n+1) + 1}{n+1} = 1 + \frac{1}{n+1},$$

we obtain

$$1 + \frac{1}{n+1} \geq \left(1 + \frac{1}{n}\right)^{\frac{n}{n+1}}.$$

Raising both sides to the power $n+1$ gives

$$\left(1 + \frac{1}{n+1}\right)^{n+1} \geq \left(1 + \frac{1}{n}\right)^n.$$

Hence (a_n) is increasing.

Step 2. (a_n) is bounded above.

Using the Binomial Theorem,

$$\begin{aligned} a_n &= \left(1 + \frac{1}{n}\right)^n \\ &= 1 + 1 + \frac{n(n-1)}{2!n^2} + \frac{n(n-1)(n-2)}{3!n^3} + \cdots + \frac{1}{n^n} \\ &< 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{(n-1)!} + \frac{1}{n!} \\ &< 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots \\ &= 3. \end{aligned}$$

Thus (a_n) is bounded above. Also, we used the fact that $k! \geq 2^{k-1}$ for every $k \geq 1$, and so we have

$$\frac{1}{k!} \leq \frac{1}{2^{k-1}}.$$

Therefore, by the Monotone Convergence Theorem, (a_n) is convergent. We denote its limit by

$$e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

The number e is one of the most important constants in mathematics.